

Humanoid Everywhere: Large-Scale Real-to-Sim Environment and Data Generator for Humanoid Loco-Manipulation

Author Names Omitted for Anonymous Review.

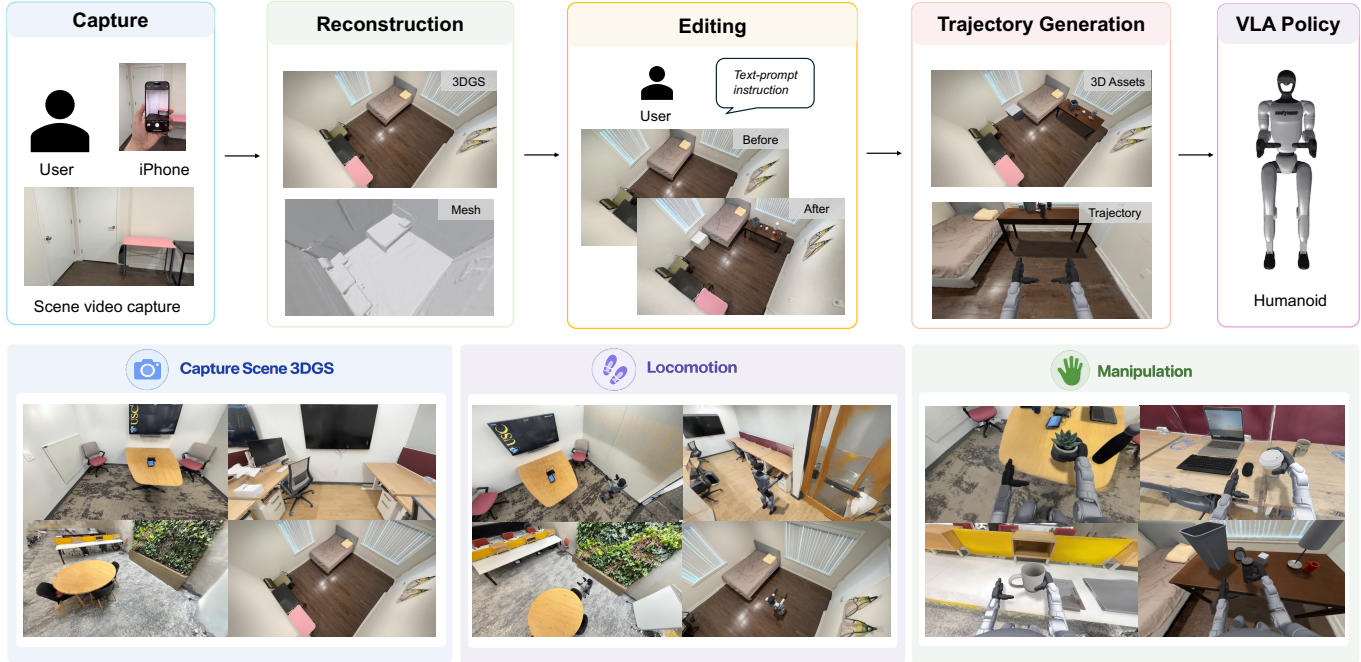


Fig. 1. **Humanoid Everywhere Data Generator.** *Top row (Our goal):* Starting from a user-captured real scene (**Capture**), we obtain captured scene 3DGS and mesh, and apply generative image editing and object-level reconstruction (**Editing**) to integrate fully composited interactive 3D assets into the environment. This asset-loaded environment is then ingested by our Gaussian simulator for physics-grounded trajectory generation (**Trajectory Generation**) to train generalizable robot policies (**VLA Policy**). *Bottom row:* Our framework scales across diverse scene captures, demonstrating promising results by generating varied locomotion and manipulation trajectories across different tasks for downstream policy training.

Abstract—A fundamental question in robot learning is *what data matters most for training generalizable humanoid policies*. Addressing this requires frameworks that balance scalability, efficiency, generality, and visual fidelity. We present **Humanoid Everywhere**, an automated Real2Sim data generator for humanoid loco-manipulation. From smartphone captures, our pipeline reconstructs photorealistic scenes via 3D Gaussian Splatting (3DGS) and populates them with foundation-model-generated interactive assets, bypassing manual modeling and per-object scanning. Central to this is a Compositional Gaussian Simulator coupling contact-rich physics with efficient rendering to enable whole-body locomotion and dexterous interaction. Guided by text instructions, this auto-generative pipeline continuously outputs fully annotated trajectories—including states, actions, and multi-view RGB-D. A per-Gaussian relighting module further multiplies visual diversity at sub-linear cost without trajectory re-simulation. By unifying real-world reconstruction, generative assets, and scalable physics, **Humanoid Everywhere** provides an open-ended data source for Vision-Language-Action (VLA)

models, establishing a data flywheel for continual real-world scaling.

I. INTRODUCTION

Robot learning lacks an “Internet for Robots”: a scalable supply of *labeled* interaction data. General-purpose humanoid manipulation remains especially bottlenecked by how data is sourced and annotated. Contemporary efforts to close this gap fall into two dominant paradigms, each facing a structural limitation: physics-based simulation and generative video world models.

The first paradigm scales data collection via physics engines. Frameworks such as InterPrior [72] and ULTRA [27] scale whole-body loco-manipulation by distilling motion data into generative controllers. While simulation yields cheap, repeatable, explicitly labeled rollouts, it remains structurally constrained: synthetic assets require unscalable manual CAD

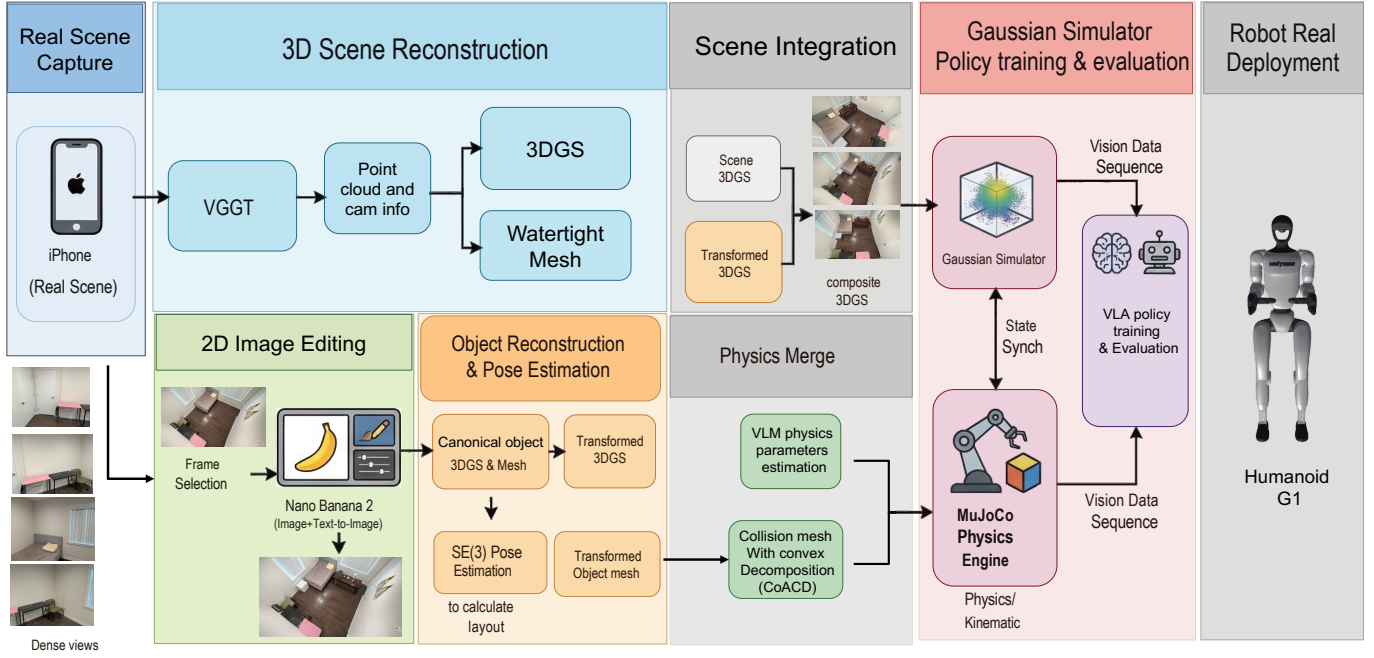


Fig. 2. **Overview of the *Humanoid Everywhere* framework.** Our system provides a scalable and efficient pipeline for robot data generation, enabling real-world scene reconstruction and the unbounded generation of manipulable 3D assets. Starting from a monocular walkthrough video, the pipeline operates in two primary stages. **(1) 3D Asset Reconstruction and Generation:** We reconstruct a photorealistic 3D Gaussian Splatting (3DGS) base scene. To populate the environment, user-provided text prompts guide 2D image editing for object insertion. These objects are extracted and converted into manipulable assets via single-image 3D reconstruction and depth-calibrated 6D pose estimation, allowing seamless integration into the base scene. **(2) Physics Integration and Gaussian Simulator:** We decompose the generated object meshes into convex collision geometries and leverage a Vision-Language Model (VLM) [2] to infer plausible physical parameters (e.g., mass, friction). Building on the SIMPLE [63] framework, we integrate these assets into our **Gaussian Simulator**, which couples a compositional 3DGS renderer with MuJoCo dynamics. This environment generates diverse contact-rich trajectories with zero-cost visual augmentation, providing synchronized, high-fidelity observations for scalable sim-to-real policy training.

authoring, and standard Domain Randomization (DR) [56] discards the subtle depth, material, and geometric cues crucial for precise real-world manipulation [12].

The second paradigm bypasses explicit simulators using generative video models. World models like Cosmos [45] and DreamZero [75] learn dynamics directly from large-scale robot videos [8, 15, 20, 74]. While visually faithful, they trade away control precision—a poor fit for contact-rich tasks. Lacking explicit physical constraints, these models suffer *physical hallucinations* (e.g., objects phasing through hands or clipping geometry) and provide no contact or joint-limit feedback. Furthermore, recovering ground-truth robot states from generated frames requires noisy post-processing, and autoregressively synthesizing high-resolution video imposes latency bounds that make large-scale rollout generation impractical. We therefore aim to fuse the photorealism of real-world capture with the rigorous control of a physics simulator.

Real2Sim via 3DGS: Bridging Both Worlds. We resolve this trade-off via a dual representation: 3D Gaussian Splatting (3DGS) [33] supplies real-world visual richness, while co-registered meshes provide the structured geometry required for accurate physical interaction. From a short smartphone video, 3DGS reconstructs a real-time-renderable scene that preserves authentic appearance and layout, sharply narrowing the visual domain gap. Explicitly decoupling this visual representation (anisotropic Gaussians) from the geometric one

(triangle meshes for collision and rigid-body control) yields both real-world appearance and physically accurate, high-throughput dynamics.

Building a seamless capture-to-simulation pipeline on 3DGS raises four challenges not jointly addressed by prior systems. **(i) Object asset creation:** 3D generative models [10] lack consistent scale and physical parameters, while 3DGS scanning is unscalable; we need a mechanism to produce arbitrary, simulation-ready assets with Gaussian appearance. **(ii) Compositional scene assembly:** Integrating disparate backgrounds, generated objects, and robots requires depth-calibrated alignment and unified transforms. **(iii) Baked illumination:** 3DGS bakes capture lighting into assets [33], causing conflicting illumination when objects move or mix. **(iv) Data diversity at scale:** While motion planners can generate stable, contact-rich rollouts [18, 38, 63], rendering them from high-quality meshes (e.g., Isaac Sim [40]) remains a computational bottleneck. This motivates a 3DGS simulator capable of re-rendering trajectories at near-zero marginal cost under varied conditions.

Our Approach. We present **Humanoid Everywhere**, an end-to-end Real2Sim framework for humanoid loco-manipulation, instantiated on a Unitree G1 humanoid (Fig. 2). From a smartphone walkthrough it produces a photorealistic 3DGS scene with a co-registered collision mesh; users populate it with novel objects via image editing and object-level reconstruction,

lifting each into a functional asset whose physical parameters and materials are inferred by foundation models [2]. For trajectory generation, we build on the SIMPLE simulator [63] and propose a Gaussian simulator that loads objects and robot links as independent Gaussian sets, each paired with an aligned collision mesh, into a compositional renderer backed by MuJoCo [57], decoupling physics from rendering for high-fidelity contact dynamics with real appearance. To resolve baked illumination, the per-object collision mesh supplies the surface normals Gaussians lack, which we transfer to apply unified Blinn–Phong shading [4] and cast shadows [66] per Gaussian. Because shading is decoupled from the physics rollout, a single trajectory is re-rendered under varied light direction, color temperature, and shadow intensity at sub-linear marginal cost. The resulting data directly supports downstream VLA policy training [3, 12, 62].

Our **contributions** are:

- **An automated, human-out-of-the-loop data generator.** We fuse real-world grounding (3DGS + mesh), web-scale foundation models (for editing [23], reconstruction [10], pose [65], and physics), and MuJoCo simulation [57]. Building on prior real-to-sim frameworks [55, 63, 76], this replaces manual teleoperation with scalable trajectory generation.
- **Multiplicative scaling with photorealistic, exact labels.** A single capture expands multiplicatively (scenes $\times$ objects $\times$ rollouts $\times$ lighting) at sub-linear marginal cost. Every frame retains real-world fidelity and outputs exact analytical labels: joint states ($\mathbf{q}_t, \dot{\mathbf{q}}_t$), SE(3) poses, and synchronized multi-view RGB-D with instance segmentation.
- **A Compositional Gaussian Simulator.** By explicitly decoupling physics from rendering, the engine populates scenes with unlimited generated assets and re-renders identical rollouts under randomized illumination. This scales visual diversity infinitely without additional capture or simulation compute.

II. RELATED WORK

Sim-to-Real for Trajectories. Physics simulators (MuJoCo [57], Isaac Sim [40], SAPIEN [68]) with asset libraries (Objaverse [14], PartNet [42]) are standard for training manipulation policies, but rely on artist-authored assets that are costly and exhibit a visual domain gap; even photorealistic RoboCasa [44] remains artist-dependent. Domain Randomization [49, 56] broadens coverage yet discards the depth and material cues precise grasping needs, while large-scale real-data policies (RT-2 [5], π_0 [3]) bypass simulation at the cost of expensive robot fleets. Real-to-sim reconstruction instead closes the visual gap at its source: NeRF [41] is too slow for interactive use, whereas 3DGS [33] enables real-time rendering and has driven robotic tracking, grasping, and SLAM [32, 39, 59] as well as physics-coupled world models [1] and editable manipulation scenes [54]. Real-to-sim-to-real pipelines build interactive digital twins (or “cousins”) from real images to

fine-tune policies before redeployment—RialTo [58], URD-Former [11], and ACDC [13]—while generative-simulation systems auto-create tasks, scenes, and assets in sim [31, 61]. Prior neural-rendering pipelines treat 3DGS as a static backdrop with naive depth compositing, and relightable 3DGS via inverse rendering [22, 29] requires slow per-scene MLPs; our per-Gaussian Blinn–Phong shading instead operates directly on SH coefficients for high-throughput generation. To populate scenes without per-object capture, we generate assets from images via diffusion-based 3D reconstruction [10, 51].

Video World Models. An alternative replaces the graphics pipeline with learned generators. From early action-conditioned video prediction [19, 20] to modern video world models [24, 73], these predict future frames and scale with data; the Cosmos family [45–47] extends this into physical-AI world foundation models with sim-to-real appearance transfer, and world-action models such as DreamZero [75] and World-VLA [8] jointly predict frames and actions. A related line treats generated video as a planner or policy—UniPi [15], Video Language Planning [16], and GR-1/GR-2 [9, 67]—or as a learned neural simulator for control [37, 70, 74, 78]. These inherit real visual complexity, but high-resolution generation is slow and physics is *implicit*, giving no guarantees on contact forces, collision geometry, or friction—a fundamental limit for contact-rich manipulation. We instead retain explicit MuJoCo physics for reliable grasp planning and whole-body control while using 3DGS for visual fidelity.

Humanoid Data Generation and Benchmarks. Scarce whole-body humanoid data motivates scalable generators, datasets, and benchmarks. One line collects human demonstrations via real-time shadowing or teleoperation [21, 25], robot-free portable capture [43], or trajectory-optimized control with teleoperated imitation [38], yielding large real-world corpora such as Humanoid Everyday [77] but keeping a human in the loop. A second line synthesizes data in simulation or pixel space: DreamGen/GR00T-Dreams [28] “dream” task videos and recover pseudo-actions to train GR00T-style policies [48], while DexMimicGen [30] scales a few demonstrations into large bimanual datasets in physics simulation. Simulated humanoid benchmarks standardize whole-body loco-manipulation evaluation—HumanoidBench [53] and Skill-Blender [35] on procedurally specified tasks, and SIMPLE [63], the loco-manipulation simulator our generator builds on. Closest in spirit, GRAIL [69] composes 3D assets with video-foundation-model priors to synthesize 4D human-object-interaction trajectories for a Unitree G1, and Gauss-Gym [17] embeds 3DGS as a drop-in renderer inside a GPU physics simulator for visual locomotion. In contrast, our generator turns a single smartphone capture into a photorealistic 3DGS real-to-sim scene with explicit MuJoCo physics, then replays relit whole-body episodes at zero marginal cost to emit fully labeled loco-manipulation data with physically consistent contacts and exact action and state labels, rather than asset-first or world-model “dreaming.”

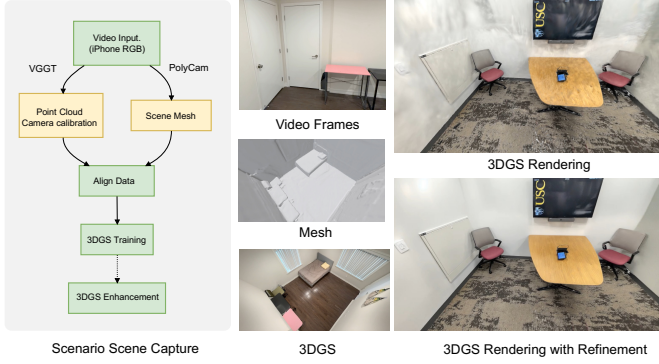


Fig. 3. **Scene capture and dual reconstruction.** From a single smartphone walkthrough, VGGT [60] extracts camera poses and a point cloud, while PolyCam [50] (or TSDF fusion) generates a geometric collision mesh. These representations are aligned to jointly initialize and train a photorealistic 3DGS scene. An optional Difx3D+ [71] refinement step (right) eliminates visual floaters and enhances rendering clarity.

III. METHOD

Fig. 2 illustrates the *Humanoid Everywhere* pipeline, which turns a single video capture of a real environment into an interactive simulation with photorealistic rendering and physics-grounded loco-manipulation. We organize it into three stages: scene capture and 3D asset generation (Sec. III-A), simulation-ready asset assembly (Sec. III-B), and a Gaussian simulator that generates and renders trajectories (Sec. III-C). Equations and implementation constants are deferred to the appendix.

A. Scene Capture and 3D Asset Generation

A user records a smartphone walkthrough of the scene—RGB frames, with LiDAR depth optional (Fig. 3). VGGT [60] jointly predicts per-frame extrinsics, intrinsics, dense depth and a point cloud, providing the geometric foundation for all later stages.

Collision mesh and mesh-initialized 3DGS. We obtain the collision mesh through two interchangeable pathways: a textured **PolyCam mesh** [50], or **TSDF fusion** of the calibrated depth (as in GaussGym [17]). The pipeline is therefore mesh-source agnostic; what manipulation needs is accurate *support surfaces* (floor, table-tops), where small errors cause unstable contact or floating objects. We fit clean planar supports from the calibrated depth and recover per-object geometry with SAM-3D [10], keeping object- and surface-level quality sufficient even when the surrounding room mesh is coarse. Using the aligned mesh as initialization, we then train a 3DGS scene [33] via *gsplat*.

Image-guided object generation. We populate the scene from a single edited image through a *segment–calibrate–reconstruct–place–validate* pipeline (Fig. 4), following OpenReal2Sim [55, 76]. From the trained 3DGS we render RGB, per-pixel depth, and intrinsics; a user edits this view with a text-to-image editor (e.g., Gemini Nano Banana [23]) to insert or rearrange objects on the support surface (e.g., “add a coffee mug and a laptop”). Each inserted object is isolated

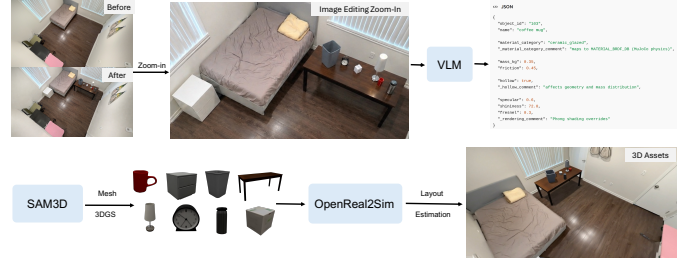


Fig. 4. **Image-guided object generation.** A rendered view is edited with a text prompt to insert objects; each is segmented and lifted to a textured mesh and 3DGS by SAM-3D, assigned physical parameters (mass, friction) by a VLM [2], and seated on the support surface by SE(3) layout optimization, yielding simulation-ready 3D assets.

as a 2D mask with SAM 2 [52] and reconstructed by SAM-3D [10] into a textured mesh and Gaussian splat. We place it on the support plane with a full 6-DoF pose, anchoring on the bottom-of-mask contact for a height-invariant initialization; FoundationPose [65] then estimates the full SE(3) pose by render-and-compare, a VLM critic (Qwen3-VL [2]) corrects residual yaw, and the transforms $\{\mathbf{T}_{\text{scene}}^{\text{obj}}\}$ form the scene layout.

B. Simulation-Ready Assets

Physical parameters and validation. The same VLM (Qwen3-VL [2]) infers each object’s simulation properties from texture, name, and size: a material category, a *volume-consistent* mass (density $\times$ reconstructed volume), and sliding/torsional/rolling friction, with a material lookup table for fallbacks. An object joins the pool only after a closed-loop check: we instantiate it in MuJoCo with its convex collision geometry and estimated mass/friction, drop it at $\mathbf{T}_{\text{scene}}^{\text{obj}}$, and run a short free-settling rollout. It is accepted when stable—resting without interpenetration, sliding, or jitter. Failures route back: implausible dynamics re-estimate mass/friction, an unstable rest pose re-runs SE(3) optimization, and severe penetration re-runs convex decomposition; persistently unstable objects are discarded.

Convex decomposition and grasp synthesis. Reconstructed meshes are non-convex, so we decompose each into convex parts with CoACD [64], applied hierarchically to background, table, and object, and referenced alongside the visual mesh in the MuJoCo XML. For each object, BODex [36] is applied for generating grasp hand pose (Unitree Dex3) and produces ~ 20 candidates (wrist SE(3) pose + 7 joint angles), and filtered for reachability from standing.

C. Gaussian Simulator for Trajectory Generation

Our backbone is 3DGS [33], which is *compositional*: backgrounds, objects, and robot links are independent Gaussian sets, rigidly transformed and composited at render time, enabling scene variation without retraining (Fig. 5). We build on the SIMPLE simulator [63], extending its MuJoCo physics and motion-planner trajectories with 3DGS rendering: each object’s Gaussian set is rigidly attached to its collision mesh;

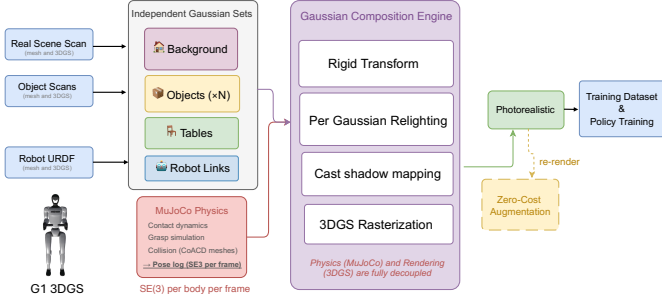


Fig. 5. **Compositional Gaussian Simulator.** The scene is represented as distinct Gaussian sets for the background, articulated robot links, and manipulable objects. MuJoCo simulates contact-rich physical dynamics, applying time-indexed SE(3) poses to transform each set rigidly. A unified illumination pipeline combines per-Gaussian relighting (albedo recovery and Blinn-Phong shading) with shadow mapping before final 3DGS rasterization. This explicit decoupling of physics and rendering allows a single trajectory rollout to be re-rendered under diverse lighting conditions at sub-linear cost.

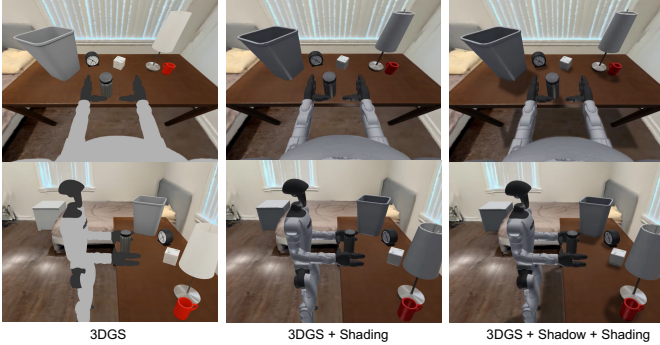


Fig. 6. **Per-Gaussian dynamic shading and cast shadows.** Naive composition (3DGS, left) leads to static, mismatched illumination that fails when assets move. Our simulator dynamically re-shades each Gaussian per frame (+Shading, middle) and integrates cast shadows (+Shadow+Shading, right) to visually ground the robot and objects. This ensures illumination coherence and enables efficient visual domain randomization by re-rendering the same trajectory under varied lighting.

per step, MuJoCo advances the state and we apply the resulting transforms before rasterization, so the solver handles contact dynamics while the rasterizer renders appearance, without modifying either.

Trajectory generation. Within this simulator, an AMO whole-body policy [26] handles bipedal balance and locomotion (legs + waist, 15 DOF), while a motion planner produces the upper-body path through waypoints `rest` $\rightarrow$ `pre_reach` $\rightarrow$ `reach_table` $\rightarrow$ `grasp` $\rightarrow$ `lift`. Every rollout is densely auto-labeled—joint states ($\mathbf{q}_t, \dot{\mathbf{q}}_t$), link and object SE(3) poses, contact forces, and synchronized multi-view observations (head-stereo and wrist RGB, depth, segmentation). Because the data is simulated, these action labels are *exact* rather than inferred.

Per-Gaussian dynamic shading. Compositing heterogeneous assets exposes a 3DGS limitation: lighting is baked in. Each Gaussian’s SH encodes its source illumination, so background, objects, and robot appear under different lights, and a grasped object’s shading no longer matches its pose (Fig. 6). We add a per-Gaussian relighting pass that modifies

SH *before* rasterization, leaving the rendering architecture unchanged. Gaussians fall in three categories: background, kept with baked SH; object, albedo recovered then re-lit; and robot, lit from a fixed metallic base color $\mathbf{a}_{\text{robot}} = [0.42, 0.44, 0.50]$. For objects with baked SH, given canonical normal $\mathbf{n}_i^{\text{can}}$ and primary light ℓ , the baked irradiance is

$$E_i^{\text{baked}} = \max(k_a + k_d \cdot \max(\mathbf{n}_i^{\text{can}} \cdot \ell, 0), \epsilon) \quad (1)$$

with k_a, k_d ambient and diffuse intensities and $\epsilon = 0.2$ preventing division-by-zero in shadow; the albedo is $\mathbf{a}_i = \text{clamp}(\mathbf{c}_i^{\text{baked}}/E_i^{\text{baked}}, 0, 1)$. We then re-shade with a modified Blinn-Phong model: for Gaussian g_i with world normal $\mathbf{n}_i$, view direction $\mathbf{v}_i$, and half-vector $\mathbf{h}_i = \text{normalize}(\ell + \mathbf{v}_i)$,

$$\mathbf{C}_i = \underbrace{\mathbf{a}_i \mathbf{c}_a k_a}_{\text{ambient}} + \underbrace{|\mathbf{n}_i \cdot \ell| \mathbf{a}_i \mathbf{c}_d k_d}_{\text{diffuse}} + \underbrace{\sum_{j=1}^2 |\mathbf{n}_i \cdot \ell_j^{\text{fill}}| \mathbf{a}_i k_j^{\text{fill}}}_{\text{fill lights}} + \underbrace{k_s (\mathbf{n}_i \cdot \mathbf{h}_i)^\alpha \mathbf{c}_\ell}_{\text{specular}} + \underbrace{k_f (1 - |\mathbf{n}_i \cdot \mathbf{v}_i|)^3 \mathbf{c}_\ell}_{\text{Fresnel}} \quad (2)$$

with per-material (k_s, α, k_f) set by the object’s VLM-inferred material category (Table I). We use two-sided Lambert $|\mathbf{n}_i \cdot \ell|$ since mesh-to-Gaussian normals may be inconsistently oriented; two opposing fill lights ℓ_j^{fill} and a floor $\mathbf{C}_i \leftarrow \max(\mathbf{C}_i, 0.38 \mathbf{a}_i)$ prevent pitch-black concavities. The shaded color becomes an SH DC coefficient via $c_i^{(0)} = (\mathbf{C}_i - 0.5)/C_0$, $C_0 = 0.28209$.

TABLE I
PER-CATEGORY BRDF PARAMETERS. EACH OBJECT REQUIRES ONLY A SINGLE MATERIAL LABEL.

Material	k_s	α	k_f
Stainless steel	0.85	100	0.40
Metal	0.80	96	0.35
Ceramic (glazed)	0.60	72	0.30
Robot (G1 Links)	0.50	48	0.20
Plastic (rigid)	0.25	28	0.12
Wood	0.10	16	0.08
Cardboard	0.05	8	0.05

Cast shadows. Without cast shadows the robot and objects appear to float; we add orthographic shadow mapping that darkens receiver surfaces under the primary light, grounding every element in the scene.

Shading augmentation. Because relighting decouples physics from rendering, the same geometric joint trajectory is *re-rendered* under different lighting (sampled on a sphere) without a new physics rollout, giving multiplicative augmentation: M rollouts $\times N$ lighting variants = $M \times N$ trajectories. We can also vary color temperature, shadow/lighting intensity as shown in Fig. 7.

Diffusion refinement and performance. 3DGS leaves blur, floaters, especially for large scene optimization with sparse views; an optional single-step Diffr3D+ [71] pass restores each frame after rasterization, compositing with our shading and shadows. It adds ~ 56 ms/frame at 320×240 (~ 9 FPS; Table IV), feasible for offline generation, with its sim-to-real effect left to future evaluation. The complete pipeline renders our ~ 2 – 5 M-Gaussian scenes (up to ~ 11 M with objects)

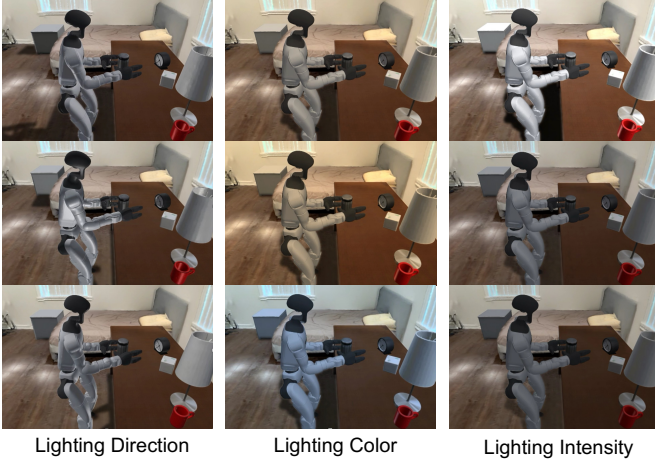


Fig. 7. **Zero-cost lighting domain randomization.** We re-render each physics rollout under randomized illumination—varying light *direction*, *color*, and *intensity* (columns)—to multiply visual diversity at marginal rendering cost, without re-simulating the trajectory (Sec. IV-C). Because the scene is captured from the real world, inserted objects stay harmonized with the background under every lighting condition, yielding realistic, fully labeled variants—unlike photorealistic synthetic-scene rendering, which is far slower.

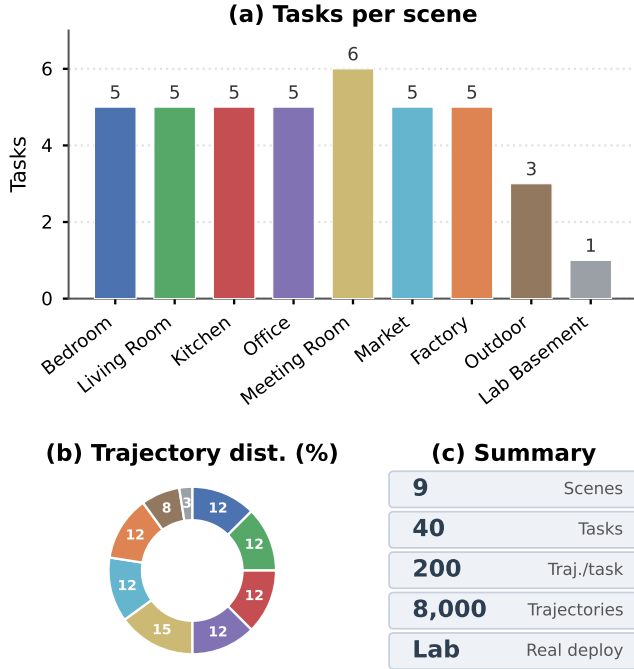


Fig. 8. Tasks per scene (a), trajectory distribution (b), and dataset summary (c).

at 16–20FPS on a single A6000 Ada GPU; Table IV and Fig. 10 (Sec. IV-C) report throughput across renderers and its dependence on Gaussian count.

IV. EXPERIMENTS

We evaluate *Humanoid Everywhere* as a data generation system along three key dimensions: **Scalability**: can it scale to many scenes and objects at low cost? **Efficiency**: how efficient is the end-to-end pipeline? **Data quality**: is the generated data sufficient to train effective policies? We claim

that *Humanoid Everywhere* is a scalable, efficient, and general-purpose Real2Sim data generator whose output is sufficient to train deployable humanoid policies. Accordingly, we organize our evaluation around these four axes: **scalability** (§IV-B), **efficiency** (§IV-C), and **data quality** (§IV-D).

A. Qualitative Results

Fig. 9 shows the generator producing feasible data end-to-end: a captured scene is reconstructed, edited and populated with generated 3D assets, then used to plan manipulation and locomotion trajectories that are replayed through the Gaussian simulator at *zero additional trajectory cost* under domain randomization—varied distractor layouts, lighting (Fig. 7), and shading—turning one rollout into many fully labeled, diverse samples. Fig. 6 shows why the replay is photorealistic: per-Gaussian shading and cast shadows give composited objects and the robot illumination-coherent appearance, unlike the baked lighting of naive 3DGS compositing.

Robot Platform and Policy Compatibility. We evaluate our framework using a Unitree G1 humanoid robot equipped with a 29-DOF upper body and 12-DOF dexterous hands. Visual observations from the robot’s head and egocentric wrist cameras are rendered. Our pipeline exports all generated trajectory data—comprising synchronized multi-view RGB sequences, proprioceptive states, and action tokens—directly into the standardized LeRobot [7] dataset format. This makes the data immediately feasible for training modern downstream control architectures, including [5, 12, 34], RT-2 [5], ensuring a seamless transition from simulation to real-world deployment.

B. Scalability

Can the pipeline scale to many scenes and objects at low marginal cost?

Adding new environments and objects is cheap in *human effort*: only three stages need the operator—a short capture, selecting which objects to reconstruct, and the text-to-image edit—while reconstruction, asset generation, physics estimation, grasp synthesis, and trajectory generation run automatically, with zero manual 3D modeling or trajectory annotation (Table II). Data collection currently supports the standard set of 9 scenes and 40 tasks across diverse environments (Fig. 8), while remaining extensible to additional scenes through flexible capture sources; Table III shows a sample of the scene-consistent objects.

C. Efficiency

How fast is end-to-end data generation?

We highlight a key advantage of 3DGS representations for trajectory replay: substantially higher rendering efficiency. As reported in Table IV, Isaac Sim’s path tracer achieves photorealistic rendering at a significantly higher computational cost (~ 0.2 FPS at 720p, corresponding to ~ 3 – 10 minutes per trajectory when scaled to 320×240). In contrast, our 3DGS rasterizer sustains 16–20 FPS at 320×240 on a single NVIDIA A6000 Ada GPU. Rendering performance scales

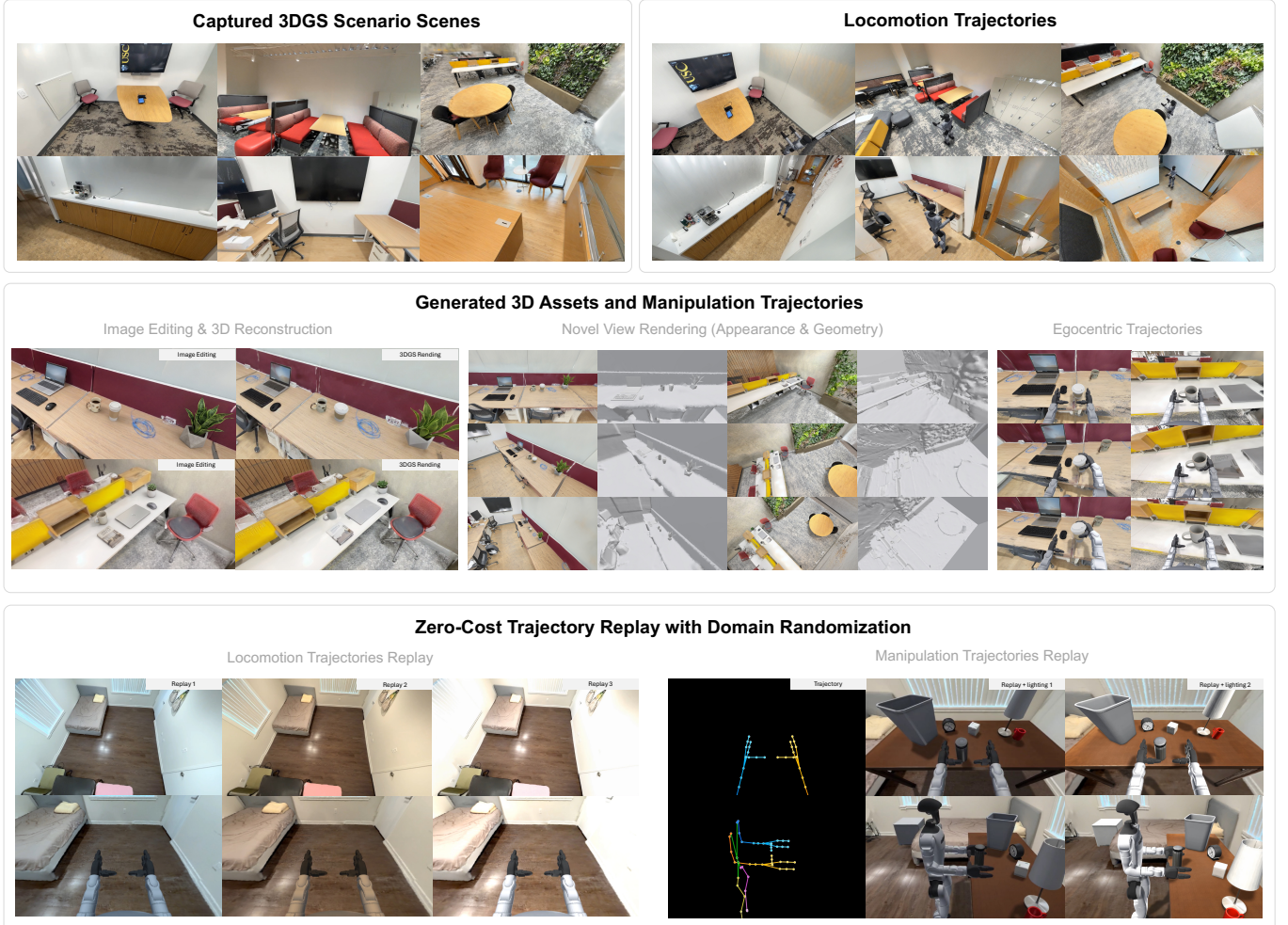


Fig. 9. **The data generator produces feasible, diverse data end-to-end.** *Top:* captured 3DGS scenes and locomotion trajectories across diverse real environments. *Middle:* from a text-edited image we generate manipulable 3D assets and physics-valid manipulation trajectories, rendered with correct novel-view appearance and geometry and egocentric head/wrist views. *Bottom:* each rollout is replayed through the Gaussian simulator at *zero additional trajectory cost* under domain randomization—varied distractor layouts, scene lighting, and shading—multiplying one trajectory into many visually diverse, fully labeled locomotion and manipulation samples.

TABLE II
PIPELINE STAGES AND AUTOMATION.

Pipeline stage	Agent / tool	Input
Scene capture	iPhone video	User
Camera calibration	VGGT	Auto
3DGS training	gsplat	Auto
Reconstruction target selection	operator picks objects	User
Collision mesh	TSDF / PolyCam + ICP	Auto
Scene editing	text-to-image	User
Object generation	SAM-3D + CoACD + Qwen3-VL	Auto
Grasp synthesis	BODex	Auto
Trajectory generation	motion planner	Auto

Only three stages require the user—scene capture, selecting which objects to reconstruct, and the text-to-image scene edit; the other six run automatically without manual annotation.

predictably with scene complexity (Fig. 10). Because rendering is decoupled from physics simulation, generating N relit variants requires only N rendering passes while reusing a single physics rollout, enabling scalable visual diversification at minimal additional computational cost.

TABLE III
SAMPLE SCENE–OBJECT COVERAGE. OBJECTS ARE DESIGNED TO BE CONSISTENT WITH EACH SCENE’S REAL-WORLD SCENARIO.

Scene	Example Objects
Bedroom	Lamp, thermos, clock, water cup
Living Room	Speaker, vase, bottles, cans
Kitchen	Orange, apple, jar, kettle, bowl
Office	Laptop, mouse, book, plant, coffee cup & mug
Meeting Room	Plant, coffee mug, marker, laptop
Market	Bottle, snack box, can, jar, fruit

A representative sample of the scene–object coverage; each scene is populated with objects consistent with its real-world scenario.

D. Data Quality

Is the generated data accurate and physically valid enough to train effective policies? Three properties make the output high-quality by construction. (i) **Exact labels:** because every rollout is simulated, joint states, per-link and object SE(3) poses, contact forces, and instance segmentation are read directly from the engine, avoiding the pose-estimation noise

TABLE IV
RENDERING THROUGHPUT ACROSS RENDERERS.

Renderer	ms/frame	FPS	Per traj.
Isaac Sim (path-traced)	~500–1700	~0.6–2	~3–10 min
Ours (3DGS)	50–62	16–20	18–22 s
Ours (3DGS + refiner)	106–118	~9	38–42 s

Performance is reported per frame and per 360-frame trajectory at 320×240 on a single NVIDIA A6000 Ada GPU. Throughput is evaluated across scenes of varying complexity. While all methods produce photorealistic renderings, our 3DGS rasterizer achieves substantially higher throughput than path-tracing-based alternatives

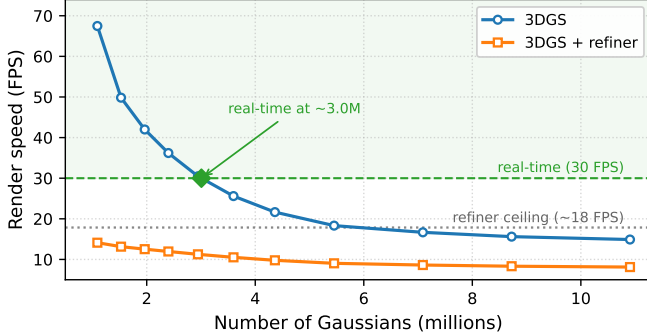


Fig. 10. **Render speed scales with the number of Gaussians** Rendering speed decreases as the number of Gaussians increases, with less complex scenes achieving real-time performance. Optional refinement further improves visual fidelity at the cost of additional computation, providing a flexible trade-off between rendering quality and efficiency

of video or human-demonstration data. **(ii) Physical validity:** each object passes a closed-loop stability check (Sec. III-B) before entering the pool, so contacts and rest poses are simulation-consistent. **(iii) Photorealism:** per-Gaussian shading and cast shadows (Fig. 6) keep composited objects and the robot illumination-coherent with the captured scene, narrowing the visual domain gap relative to synthetic-mesh rendering. Quantifying downstream policy success on hardware is left to future work (Sec. V).

E. Discussion: What Data Do Robots Really Need?

Balancing scale and fidelity. The defining tension in robot data is scale versus fidelity. Teleoperation gives high-fidelity, exactly-labeled demonstrations but scales only with human effort; human video and wearables are abundant but lack action labels and physical grounding; conventional simulation is endlessly scalable yet visually impoverished. Our Real2Sim generator does not pick a side—it dissolves the trade-off: one real capture supplies real-world visual and physical grounding, while the compositional Gaussian simulator multiplies it into an open-ended stream of rollouts at simulation scale, each carrying *exact* action, state, and contact labels rather than inferred ones. The same engine fuses the three complementary data sources—web-scale semantic priors (foundation models for editing, segmentation, reconstruction, and physical-parameter inference), simulation scale (3DGS–MuJoCo), and real-world grounding (3DGS capture)—and, because new captures and prompts regenerate data exactly where a deployed policy is



Fig. 11. **A generative path to unbounded scaling.** Real capture is not required: a single image—captured *or* generated—is expanded by a video world model (e.g., Genie 3 [6]) into a full 3DGS scene (*bottom*), then populated with unlimited generated objects in varied layouts (*top*) and loaded into the simulator for trajectory generation, scaling data with no additional user effort.

weakest, naturally closes the loop as a post-deployment data flywheel.

Toward unbounded scaling. From our experiment, we also find that user capture is not even strictly required (Fig. 11): an input image—captured *or* generated—can be expanded by a video world model [6] into a complete 3DGS scene and populated with unlimited generated objects in varied layouts, then loaded into the simulator for trajectory generation. This removes per-scene capture effort and points toward generating fully labeled data without bound.

V. CONCLUSION

Summary. We presented **Humanoid Everywhere**, a data generator that transforms a single smartphone capture into an open-ended, fully labeled supply of contact-rich humanoid loco-manipulation data. By unifying 3DGS real-world grounding, web-scale foundation models, and a compositional 3DGS–MuJoCo simulator, it resolves the scale-fidelity trade-off. It eliminates manual CAD and per-object scanning, provides exact analytical action labels, and leverages per-Gaussian dynamic shading to multiply visual diversity under varied illumination at sub-linear marginal cost.

Future Work. Immediate next steps focus on scaling dataset generation and validating real-to-sim-to-real humanoid real deployment. A key objective is closing the loop: re-capturing post-deployment failures and editing them into targeted corrective scenes to establish a continuous data flywheel. Long-term extensions include multi-room mobile manipulation, and no-rigid object manipulation. By requiring only a smartphone and a text prompt to mint fully labeled simulation worlds, this pipeline takes a practical step toward an accessible “Internet for Robots.”

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